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Eye drop dispenser

Abstract

A dispenser assembly includes a dispenser bottle having a nipple for dispensing a solution (e.g., eye drop solution, eye wash solution, eye lid cleaner and/or wash solution), and a cap removably couplable to the dispenser bottle over the nipple. The dispenser bottle has a chamber that removably receives a single-use vial cartridge. An eye cup is fixedly coupled to the dispenser bottle around the nipple, or the eye cup is detachable and invertible relative to the dispenser bottle. The eye cup can be supported on or rest on the periphery of the eye socket to facilitate delivery of the solution onto or over a user's eye.

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS (1) Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57. This application is a continuation-in-part of U.S. patent application Ser. No. 18/654,342, filed on May 3, 2024, the entire disclosure of which is incorporated herein by reference.

FIELD

(1) The present disclosure is directed to an eye drop or eye wash or eye lid cleaner dispenser bottle, and more particularly to an eye drop or eye wash or eye lid cleaner dispenser with an eye cup for dispensing eye drops or eye wash or eye lid cleaner.

DESCRIPTION OF THE RELATED ART

(2) Eye drops are regularly used to treat eye conditions such as dry eye, bacterial infections, eye injury, or eye conditions such as glaucoma and cataracts. Existing eye drop dispensers are deficient in that they do not accurately or consistently deliver the drops into the eye (e.g., affecting children and elderly people more), often delivering the drops on other parts of the user's face, leading to multiple attempts and making the solution difficult to administer. These multiple attempts cause an unnecessary wasting of the eye drops, thereby costing the consumer more. Another deficiency of existing eye drop dispensers is that user's may contaminate existing eye drop dispensers by touching their eye or face with the dispenser causing bacteria to grow on the dispenser tip and leading to eye infections. When the user suffers from hand tremors (e.g., essential tremor disorder or other neurological conditions) which cause shaking hands while trying to deliver the eye drops, the tremors exacerbate the defects of existing eye drop dispensers. Because existing dispenser designs make delivering eyedrops difficult, there is a decrease in compliance by users to take their medication.

SUMMARY

(3) The systems, methods, and devices described herein have innovative aspects, no single one of which is indispensable or solely responsible for their desirable attributes. Without limiting the scope of the claims, some of the advantageous features will now be summarized.

(4) In some aspects, the techniques described herein relate to an eye drop dispenser assembly, including: a dispenser bottle having a circumferential sidewall, a first surface, and a chamber extending between an opening in the first surface and a bottom of the dispenser bottle. The dispenser assembly further including an eye cup fixedly coupled to a second surface of the dispenser bottle, the eye cup having a circumferential sidewall disposed around the chamber, and a distal edge of the circumferential sidewall defining an oval shape, the distal edge configured to be disposed over an eye socket to facilitate delivery of a solution into a user's eye.

(5) In some aspects, the dispenser assembly further including a cap removably couplable to the dispenser bottle over the chamber.

(6) In some configurations, the dispenser bottle also has a shoulder, and the cap is removably couplable to the shoulder.

(7) In some configurations, an outer surface of the cap is tangential to the dispenser bottle.

(8) In some configurations, the dispenser bottle has a recess cavity extending away from the distal edge of the eye cup.

(9) In some configurations, the circumferential wall of the eye cup is partially disposed within the recess cavity.

(10) In some aspects, the techniques described herein relate to an eye drop dispenser assembly, including: a dispenser bottle including a circumferential sidewall, a first surface, and a chamber extending between an opening in the first surface and a bottom of the dispenser bottle. The dispenser assembly further including: a vial having a nipple, wherein the vial is removably disposed within the chamber, and an eye cup fixedly coupled to a second surface of the dispenser bottle, the eye cup having a circumferential sidewall disposed around the chamber, and a distal edge of the circumferential wall defining an oval shape, the distal edge configured to be disposed over an eye socket to facilitate delivery of a solution into a user's eye.

(11) In some configurations, the nipple is aligned with a central axis of eye cup.

(12) In some configurations, the vial is a single-use vial cartridge.

(13) In some configurations, the dispenser assembly has a retention lip extending from the

chamber, the retention lip configured to retain the vial in the chamber.

(14) In some configurations, the nipple does not extend past the second surface.

(15) In some configurations, a maximum width and depth of the eye cup is smaller than a maximum width and depth of the dispenser bottle.

(16) The eye cup inhibits the nipple from contacting an eye.

(17) In some configurations, the distal edge of the eye cup is contoured.

(18) In some configurations, the dispenser assembly further includes a cap removably couplable to the dispenser bottle over the nipple, the cap having an outer wall, an inner wall extending within the outer wall, and a seal at or proximal an end of the inner wall.

(19) In some configurations, the inner wall seals the nipple substantially simultaneously with the outer wall coupling to a shoulder of the dispenser bottle.

(20) In some aspects, the techniques described herein relate to an eye drop dispenser assembly, including: a dispenser bottle having a chamber, a coupling mechanism proximate the chamber, a vial having a nipple, the vial removably disposed within the internal chamber, and an eye cup having a first open end and a second open end opposite the first open end and configured to removably fit over at least a portion of the dispenser bottle, the first open end having an oval shape and the second open end having a circular shape, the eye cup being couplable to the coupling mechanism of the dispenser bottle in a stowed position where the eye cup is disposed below the nipple and about the upper portion of the dispenser bottle, the eye cup further being couplable to the coupling mechanism of the dispenser bottle in a deployed position where the eye cup is disposed so that it extends around and distally from the nipple, wherein the eye cup in the deployed position facilitates delivery of eye drops into a user's eye.

(21) In some configurations, the dispenser bottle further comprises an upper portion recessed relative to a lower portion, and wherein an outer surface of the eye cup circumferentially aligns with an outer surface of the lower portion of the dispenser bottle when in the stowed position.

(22) In some configurations, the dispenser bottle has an upper portion between the lower portion and the coupling mechanism, the lower portion having a greater width than the upper portion, the upper portion tapering from the lower portion toward the nipple.

(23) In some configurations, the dispenser bottle further comprises a bottom surface having an opening into the chamber.

(24) In some configurations, the vial is insertable into the chamber via the opening in the bottom surface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a schematic front view of one embodiment of an eye drop dispenser assembly.

(2) FIG. 1B is a schematic front view of a second embodiment of an eye drop dispenser assembly.

(3) FIG. 2A is a schematic front view of the eye drop dispenser assembly of FIG. 1A with its cap removed and spaced above the dispenser bottle, the dashed line showing the outline of the cap when covering the eye cup.

(4) FIG. 2B is a schematic side view of the eye drop dispenser assembly of FIG. 1A with its cap removed and spaced above the dispenser bottle.

(5) FIG. 2C is a schematic cross-sectional view of the eye drop dispenser assembly of FIG. 1A, where the section line is through the midplane of the side view in FIG. 2B.

(6) FIG. 3A is a schematic front view of the eye drop dispenser assembly of FIG. 1B with its cap removed and spaced above the dispenser bottle, the dashed line showing the outline of the cap when covering the eye cup.

(7) FIG. 3B is a schematic side view of the eye drop dispenser assembly of FIG. 1B with its cap

removed and spaced above the dispenser bottle.

(8) FIG. 3C is a schematic cross-sectional view of the eye drop dispenser assembly of FIG. 1B, where the section line is through the midplane of the side view in FIG. 3B.

(9) FIG. 4A is a schematic top view of the eye drop dispenser assembly of FIG. 1A.

(10) FIG. 4B is a schematic top view of the eye drop dispenser assembly of FIG. 1B.

(11) FIG. 5A is a schematic top view of the cap of the eye drop dispenser assembly of FIG. 1A and FIG. 1B.

(12) FIG. 5B is a schematic bottom view of the cap of FIG. 5A.

(13) FIG. 5C is a schematic cross-sectional view of the front view of one embodiment of the cap of FIG. 5A, where the section line is through the midplane of the side view.

(14) FIG. 5D is a schematic cross-sectional view of the front view of a second embodiment of the cap of FIG. 5A, where the section line is through the midplane of the side view.

(15) FIG. 6A is a schematic view of the eye drop dispenser of FIG. 1A in an inverted position over a user's eye to deliver an eye drop thereto.

(16) FIG. 6B is a schematic view of the eye drop dispenser of FIG. 1B in an inverted position over a user's eye to deliver an eye drop thereto.

(17) FIG. 7A is a schematic cross-sectional view of the front view of a third embodiment of the eye drop dispenser assembly.

(18) FIG. 7B is a schematic cross-sectional view of the front view of the second embodiment of the eye drop dispenser assembly.

(19) FIG. 8A is a schematic top view of the eye drop dispenser assembly of FIG. 7A.

(20) FIG. 8B is a schematic top view of the eye drop dispenser assembly of FIG. 7B.

(21) FIG. 8C is a schematic bottom view of the eye drop dispenser assembly of FIG. 7A.

(22) FIG. 9A is an assembled front view of a fourth embodiment of an eye drop dispenser assembly.

(23) FIG. 9B is an exploded view of the eye drop dispenser assembly of FIG. 9A.

(24) FIG. 9C is a schematic cross-sectional front view of the eye drop dispenser assembly of FIG. 9A.

(25) FIG. 10 is a schematic cross-sectional front view of the eye drop dispenser assembly of FIG. 9A with its cap removed and the eye cup in a deployed position extending around a nipple of the dispenser bottle.

(26) FIG. 11 is another schematic cross-sectional front view of the eye drop dispenser assembly of FIG. 9A.

(27) FIG. 12 is another schematic cross-sectional front view of the eye drop dispenser assembly of FIG. 9A with its cap removed and the eye cup in a deployed position extending around a nipple of the dispenser bottle.

(28) FIG. 13 is a schematic top view of the eye drop dispenser assembly of FIG. 9A with the cap removed.

(29) FIG. 14 is a schematic bottom view of the eye drop dispenser assembly of FIG. 9A.

(30) FIG. 15A is a schematic view of the eye drop dispenser of FIG. 9A illustrating the insertion of the single-use vial cartridge into the eye drop dispenser bottle.

(31) FIG. 15B is a schematic view of the eye drop dispenser of FIG. 9A in an inverted position over a user's eye to deliver an eye drop thereto.

(32) FIG. 16A is a schematic view of the eye drop dispenser of FIG. 7A illustrating the insertion of the single-use vial cartridge into the eye drop dispenser bottle.

(33) FIG. 16B is a schematic view of the eye drop dispenser of FIG. 7A in an inverted position over a user's eye to deliver an eye drop thereto.

(34) FIG. 16C is a schematic cross-sectional view of the eye drop dispenser in FIG. 16A.

(35) FIG. 17 is a schematic front view of a single-use vial cartridge.

(36) FIG. 18A is a schematic front cross-sectional view of a dispenser assembly.

(37) FIG. 18B is a schematic side cross-sectional view of the dispenser assembly in FIG. 18A.

(38) FIG. 19A is a schematic front cross-sectional view of a dispenser assembly.

(39) FIG. 19B is a schematic side cross-sectional view of the dispenser assembly in FIG. 19A.

DETAILED DESCRIPTION

(40) Embodiments of the eye drop dispenser assembly will now be described with reference to the accompanying figures, wherein like numerals refer to like or similar elements throughout. Although several embodiments, examples, and illustrations are disclosed below, it will be understood by those of ordinary skill in the art that the inventions described herein extends beyond the specifically disclosed embodiments, examples, and illustrations, and can include other uses of the inventions and obvious modifications and equivalents thereof. The terminology used in the description presented herein is not intended to be interpreted in any limited or restrictive manner simply because it is being used in conjunction with a detailed description of certain specific embodiments of the inventions. In addition, embodiments of the inventions can comprise several novel features and no single feature is solely responsible for its desirable attributes or is essential to practicing the inventions herein described.

(41) FIGS. 1A, 2A-2C, 4A, 5C and 6A show a dispenser assembly **100** (the “assembly **100**”). The assembly **100** includes a dispenser bottle **150** (e.g., dispenser body), an eye cup **140**, a nipple **130**, and a cap **110**. The dispenser bottle **150**, the eye cup **140**, and/or the cap **110** can optionally be made in part or whole of plastic (e.g., polypropylene), silicone or rubber. In one example, the eye cup **140** can be translucent. In one example, the eye cup **140** can be transparent. However, the dispenser bottle **150**, the eye cup **140**, and/or the cap **110** can be made of other suitable materials (e.g., flexible or resilient materials). The eye cup **140** may be formed from a clear or transparent material. The dispenser assembly **100** can house and deliver a solution (e.g., eye drops, eye wash, eye lid cleaner and/or wash) onto or over the eye or a portion thereof (e.g., eye lid).

(42) FIGS. 1B, 3A-3C, 4B, 5D, 6B, 7B and 8B show views of a dispenser assembly **100'** (hereafter “the assembly **100'**”), which is a compact (e.g., small or pocket sized) version of the dispenser assembly **100**. The compact dispenser assembly **100'** is also referred to as the multi-use assembly **100'**. Some of the features of the assembly **100'** are similar to features of the assembly **100** in FIGS. 1A, 2A-2C, 4A, 5C and 6A. Thus, reference numerals used to designate the various components of the assembly **100'** are identical to those used for identifying the corresponding components of the assembly **100** in FIGS. 1A, 2A-2C, 4A, 5C and 6A, except that a “'” has been added to the numerical identifier. Therefore, the structure and description for the various features of the assembly **100** and how it is operated and controlled in FIGS. 1A, 2A-2C, 4A, 5C and 6A are understood to also apply to the corresponding features of the assembly **100'**, except as described below. The dispenser assembly **100'** can house and deliver a solution (e.g., eye drops, eye wash, eye lid cleaner and/or wash) onto or over the eye or a portion thereof (e.g., eye lid).

(43) The dispenser assembly **100** includes a dispenser bottle **150** with a base **152** and a cap **110** removably couplable to an end of the dispenser bottle **150** opposite to (e.g., distal from) the base **152**. In some embodiments, the base **152** is flat. In other embodiments, a portion, but less than all, of the base **152** is flat. The base **152** allows the dispenser bottle **150** to stand upright when not in use which advantageously reduces the amount of fluid (e.g., eye drops, eye wash, eye lid cleaner and/or wash) wasted. The cap **110** is removably coupled to the dispenser bottle **150**. The assembly **100** has a height A measured from the base **152** to a top of the cap **110**. In one implementation, the height A is between about 2 and 3 inches, such as 2.5 inches.

(44) The dispenser assembly **100'** includes a dispenser bottle **150'** (e.g., dispenser body) with a base **152'** and a cap **110'** removably couplable to an end of the dispenser bottle **150'** opposite to (e.g., distal from) the base **152'**. In some embodiments, the base **152'** is flat. In other embodiments, a portion, but less than all, of the base **152'** is flat. The assembly **100'** has a height A' measured from the base **152'** to a top of the cap **110'**. In one implementation, the height A' is between about 1.5 and 2.5 inches, such as 2 inches. In one example, the eye cup **140'** can be translucent. In one example,

the eye cup **140'** can be transparent.

(45) FIG. 2A depicts a front view of the dispenser assembly **100** shown in FIG. 1A with the cap **110** removed. The dispenser bottle **150** has the base **152** connected to a side wall **158**. The side wall **158** is circumferential and extends from the base **152**. The dispenser bottle **150** has a nipple **130** projecting from a surface **164** of the dispenser bottle **150** that is at an opposite end of the dispenser bottle **150** (e.g., distal) from the base **152**. The nipple **130** can be centered relative to the dispenser bottle **150**. The nipple **130** can have a circular profile, as shown for example in FIG. 4A; however, the nipple **130** can have other suitable shapes. In some embodiments, the circumferential side wall **158** can flex (e.g., when the dispenser bottle **150** is squeezed). The dispenser bottle **150** also has a shoulder **154** that circumscribes a second surface **160**. The second surface **160** connect to the circumferential side wall **158**. In the illustrated embodiment the surface **164** is coplanar with the second surface **160**. The cap **110** is removably couplable to the dispenser bottle **150** around the shoulder **154**. In some examples the cap **110** is removably couplable to the shoulder **154** with an annular snap-fit. In some embodiments, the shoulder **154** is disposed to extend above the second surface **160**. In other embodiments, the shoulder **154** is formed integral to the second surface **160** (e.g., the second surface **160** forms the upper surface of the shoulder **154** such that when the cap **110** is on, or connects to, the second surface **160** fits within the cap **110** rather than the cap **110** sitting on the second surface **160**). In the illustrated arrangement, the assembly **100** also includes an eye cup **140** integrated with the dispenser bottle **150**, which provides a supporting structure to the dispenser bottle **150** when it is inverted for delivery of solution on or over the eye (e.g., the eye cup **140** resting on the user's eye socket to support the dispenser bottle **150** in the inverted position). The eye cup **140** is coupled to the second surface **160** of the dispenser bottle **150** (e.g., to an end of the dispenser bottle **150** opposite the base **152**). The eye cup **140** can fit within the cap **110** when the cap **110** is coupled to the dispenser bottle **150**. In some embodiments, the eye cup **140** is fixedly coupled to the second surface **160** (e.g., so that the eye cup **140** is integral to, and cannot be detached or removed from, the dispenser bottle **150**). The eye cup **140** has a circumferential wall **142** that extends to an edge **144** that is distal to the dispenser bottle **150**. In one example, the eye cup **140** (e.g., the circumferential wall **142**) tapers toward the second surface **160**. The circumferential wall **142** defines an oval shape surrounding the nipple **130** (e.g., when the dispenser assembly **100** is viewed from the top with the cap **110** removed, see FIG. 4A. The circumferential wall **142** surrounds the nipple **130**, and the nipple **130** is disposed between the distal edge **144** of the circumferential wall and the second surface **160** of the dispenser bottle **150**.

(46) The dispenser bottle **150** can optionally be made of plastic, silicone, rubber, or other suitable resilient or flexible material. The dispenser bottle **150** can optionally be a single piece (e.g., a monolithic, seamless single piece). In some embodiments, the base **152**, side wall **158**, surface **164**, second surface **160**, shoulder **154**, nipple **130**, and eye cup **140**, or any combination thereof, may be formed as a single piece. The single piece can be an integrated assembly of several parts or can be a unitary structure formed from a single piece of material. In some embodiments, the base **152**, side wall **158**, surface **164**, second surface **160**, shoulder **154**, nipple **130**, and eye cup **140** may be made from plastic, rubber, silicone or any suitable material. Any of these portions may be made from the same material or from different materials.

(47) In the illustrated arrangement, the assembly **100** has a width C. In one implementation, the width C is between 1.5 and 2.5 inches, such as 2 inches. The dispenser bottle **150** has a height B measured from the base **152** to the surface **164**. In one implementation, the height B is between 1.25 and 2 inches, such as 1.625 inches. The dispenser bottle **150** also has a height I measured from the surface **164** to the top of the nipple **130**. The assembly **100** has a height F measured from the second surface **160** to the distal edge **144**. Height F is advantageously larger than height I to inhibit (e.g., prevent) the nipple **130** from contacting the eye **300** when in use. This positioning advantageously prevents (e.g., inhibits) contamination of the nipple **130** or injury to the eye **300** during use. Additionally, in some examples the height I may be less than half of height F which

advantageously further reduces the chance that the nipple **130** will inadvertently touch the user's eye **300** or otherwise be contaminated. In other embodiments, the height I may be greater than half the height F. In one implementation, the height I is between 0 and 0.4 inches, such as 0.2 inches and the height F is between 0.25 and 0.75 inches, such as 0.5 inches. The assembly **100** also has a height D. Height D is the height of the cap **110**. In one implementation, height D is greater than height F. In one implementation, the height D is between 0.5 and 1.0 inches, such as 0.875 inches. (48) With reference to FIG. 2B, the assembly **100** has a depth G. The base **152** has a depth H. Depth G may be larger than depth H. In one implementation, the depth G is between 1 and 1.5 inches, such as 1.25 inches and the depth H is between 0.5 and 1 inch, such as 0.75 inches. The eye cup **140** has a depth J. Depth J may be smaller than Depth G. In one implementation, the depth J is between 1 and 1.25 inches, such as 1.1 inches. The cap **110** has a depth K (at its distal end). In one implementation, the depth K is between 0.5 and 0.75 inches, such as 0.7 inches.

(49) FIG. 2C depicts a cross-sectional view of the dispenser assembly **100** shown in FIG. 2A. The section is taken at a midpoint of the depth G of the assembly **100** (see FIG. 2B). The dispenser bottle **150** has a chamber **162** that can hold a liquid solution (e.g., eye drop solution, eye wash solution, eye lid cleaner or wash solution). The cap **110** is connected to the dispenser bottle **150**. The eye cup **140** fits within a cavity **112** of the cap **110** when the cap **110** is removably coupled to the dispenser bottle **150**. The cap **110** has an outer wall **114** that defines the cavity **112**. Within the cavity **112**, is a post **118** extending from an underside of the outer wall **114**. In one example, the post **118** is solid. In another example, the post **118** includes a hollow portion (e.g., at a distal end of the post **118**). The post **118** has a seal **120** at its distal end. In one example, the seal **120** is separate from the post **118** and is disposed within the hollow portion of the post **118**. In another example, the seal **120** is separate from the post **118** and is attached to an end of the post **118**. In another example, the seal **120** is integral (e.g., one piece, monolithic) with the post **118** (e.g., the seal **120** is a distal portion of the post **118** and made of the same or similar material as the post **118**). When the cap **110** is coupled to the dispenser bottle **150**, the outer wall **114** is coupled to (e.g., disposed around) the shoulder **154** of the dispenser bottle **150**. The post **118** and the seal **120** define an inner wall **116** that is coupled to (e.g., disposed around) the nipple **130** so that a distal portion of the post **118** encircles the nipple **130** and is coupled to the nipple **130**. The seal **120** sits adjacent the top of the nipple **130** and presses against the nipple **130** to inhibit (e.g., prevent) leakage of solution from the nipple **130**. In another example, when the cap **110** is coupled to the dispenser bottle **150**, the seal **120** at the end of the post **118** engages (e.g., contacts) the nipple **130** to inhibit (e.g., prevent) leakage of solution from the dispenser bottle **150**. In one example, the seal **120** can have a shape similar to (e.g., a mirror image of) a shape of the nipple **130**). The post **118** is disposed within the eye cup **140** when the cap **110** is connected to the dispenser bottle **150** (e.g., when the cap **110** is coupled to the dispenser bottle **150** the post **118** is surrounded by the eye cup **140**, which is surrounded by the outer wall **114**).

(50) FIG. 3A depicts a front view of the dispenser assembly **100'** shown in FIG. 1B with the cap **110'** removed. The assembly **100'** has a dispenser bottle **150** having a base **152'** connected to a side wall **158'**. The dispenser bottle **150'** has a nipple **130'** projecting from a surface **164'** (e.g., a recessed surface shown in FIG. 3C) of the dispenser bottle **150'** that is at an opposite end of the dispenser bottle **150'** (e.g., distal) from a base **152'**. The nipple **130'** can have a circular profile, as shown for example in FIG. 4B; however, the nipple **130'** can have other suitable shapes. The dispenser bottle **150'** also has a shoulder **154'** circumferential to a second surface **160'** and distal to the base **152'**, as shown in FIG. 3A. The side wall **158'** also connected to the second surface **160'**. In some embodiments, the shoulder **154'** is disposed to extend above the second surface **160'**. In other embodiments, the shoulder **154'** is formed integral to the second surface **160'** (e.g., the second surface **160'** forms the upper surface of the shoulder **154'** such that when the cap **110'** is on, the second surface **160'** fits within the cap **110'** rather than the cap **110'** sitting on the second surface **160'**). The second surface **160'** is parallel to the base **152'**. In the illustrated arrangement, shown in

FIG. 3C, the assembly **100'** also includes an eye cup **140'**. The eye cup **140'** is fixedly coupled or integrated to the dispenser bottle **150'**. When the cap **110'** is coupled to the dispenser bottle **150'**, the eye cup **140'** can fit within the cap **110'**. The eye cup **140'** has a circumferential wall **142'** and an edge **144'** (e.g., a top edge) that is distal to the dispenser bottle **150'**. In one example, the eye cup **140'** (e.g., the circumferential wall **142'**) tapers toward the second surface **160'**.

(51) With reference to FIG. 3A, the assembly **100'** has a width C'. In one implementation, the width C' is between 1.5 and 2.5 inches, such as 1.9 inches.

(52) With reference to FIG. 3B, the assembly **100'** has a depth G'. The base **152'** has a depth H'. Depth G' may be larger than depth H'. In one implementation, the depth G' is between 1 and 1.75 inches, such as 1.2 inches or 1.4 inches and the depth H' is between 0.5 and 1 inches, such as 0.8 inches. The eye cup **140'** has a depth J'. Depth J' may be smaller than Depth G'. In one implementation, the depth J' is between 1 and 1.5 inches, such as 1.1 inches or 1.25 inches. The cap **110'** has a depth K' (at its distal end). In one implementation, the depth I' is between 0.5 and 1 inches, such as 0.9 inches or 0.75 inches.

(53) FIG. 3C depicts a cross-sectional view of the dispenser assembly **100'** shown in FIG. 3A. The section is taken at a midpoint of the depth G' of the assembly **100'** (see FIG. 3B). The dispenser bottle **150'** has a chamber **162'** that can hold a liquid. The cap **110'** is connected to the dispenser bottle **150'**. The dispenser bottle **150'** has a nipple **130'** projecting from a surface **164'**. The surface **164'** is between the second surface **160'** and the base **152'**. The positioning of the surface **164'** creates a recess cavity **156'** around the nipple **130'**. The recess cavity **156'** extends into the dispenser bottle **150'** and away from the distal edge **144'** of the eye cup **140'**. The positioning of the recess cavity **156'** allows the nipple **130'** to be recessed proximal (e.g., the nipple **130'** is nested in the recess cavity **156'** of the dispenser bottle **150'**) to the dispenser bottle **150'** while projecting towards the distal edge **144'** of the eye cup **140'**, allowing the dispenser assembly **100'** to be more compact than the dispenser assembly **100**. The eye cup **140'** fits within a cavity **112'** of the cap **110'** and the circumferential wall **142'** defines an oval shape surrounding the nipple **130'**. The nipple **130'** is disposed between distal edge **144'** of the circumferential wall **142'** and the surface **164'** of the dispenser bottle **150'**. In one example, the circumferential wall **142'** is partially disposed within the recess cavity **156'**. The cap **110'** has an outer wall **114'** that defines the cavity **112'**. Within the cavity **112'**, is a post **118'** extending from the outer wall **114'**. In one example, the post **118'** is solid. In another example, the post **118'** includes a hollow portion (e.g., at a distal end of the post **118'**). The post **118'** has a seal **120'** at its distal end. In one example, the seal **120'** is separate from the post **118'** and is disposed within the hollow portion (e.g., distal hollow portion) of the post **118'**. In another example, the seal **120'** is separate from the post **118'** and is attached to an end of the post **118'**. In another example, the seal **120'** is integral (e.g., one piece, monolithic) with the post **118'** (e.g., the seal **120'** is a distal portion of the post **118'** and made of the same or similar material as the post **118'**). The outer wall **114'** is disposed around the shoulder **154'** of the bottle **150'**. The post **118'** and the seal **120'** define an inner wall **116'**. The seal **120'** sits adjacent the top of the nipple **130'** and presses against the nipple **130'** to inhibit leakage of a solution or liquid from the nipple **130'**. In one example, the seal **120'** can have a shape similar to (e.g., a mirror image of) a shape of the nipple **130'**. The post **118'** is disposed within the eye cup **140'** when the cap **110'** is connected to the dispenser bottle **150'**.

(54) In the illustrated arrangement, the bottle **150'** has a height B' measured from the base **152'** to the second surface **160'**. In one implementation, the height B' is between 1 and 1.75 inches, such as 1.25 inches or 1.5 inches. The assembly **100'** has a height F' (for the eye cup **140'**) measured from the surface **164'** to the distal edge **144'**. In one example, the eye cup **140'** extends from the second surface **160'** (e.g., the eye cup **140'** is outside the recess cavity **156'**). In another example, the eye cup **140'** extends from the surface **164'** so that at least a portion of the circumferential wall **142'** of the eye cup **140'** is in the recess cavity **156'** of the bottle **150'**. In one implementation, the height F' is between 0.25 and 0.75 inches, such as 0.4 inches or 0.5 inches. The assembly **100'** also has a

height D'. Height D' is the height of the cap **110'**. The height D' is greater than height F'. In one implementation, the height D' is between 0.5 and 1 inches, such as 0.75 inches or 0.9 inches. The dispenser bottle **150'** has a height E' measured from the second surface **160'** to the surface **164'** (e.g., the bottom surface of the recess cavity **156'**). In one implementation, the height E' is between 0.1 and 0.6 inches, such as 0.25 inches or 0.6 inches. In some arrangements, the second surface **160'** may be coplanar with the surface **164'** and height E' is zero inches. In some examples, the second surface **160'** and the surface **164'** can be portions of the same surface.

(55) FIG. 4A depicts a top view of the assembly **100**. The dispenser bottle **150** is shown with the distal edge **144** of the eye cup **140** having an oval shape and disposed around the nipple **130**. In this embodiment, the dispenser bottle **150**, has an oval shaped perimeter. In other implementations, the perimeter of the dispenser bottle **150** or eye cup **140** may take different forms (e.g. substantially polygonal, or substantially circular).

(56) FIG. 4B depicts a top view of the assembly **100'**. The dispenser bottle **150'** is shown with the distal edge **144'** of the eye cup **140'** having an oval shape and disposed around the nipple **130'**. The nipple **130'** is centered relative to the eye cup **140'** and the dispenser bottle **150'**. In this embodiment, the dispenser bottle **150'**, has an oval shaped perimeter. In other implementations, the perimeter of the dispenser bottle **150'** or eye cup **140'** may take different forms (e.g. substantially polygonal, or substantially circular).

(57) FIGS. 5A-5D depict the cap **110**. FIG. 5A depicts a top view of the cap **110** and FIG. 5B depicts a bottom view of the cap **110**. It is understood that this illustration can be either of the embodiments shown in FIGS. 1A and 1B. FIG. 5C depicts a cross-sectional front view of one embodiment of the cap **110** and FIG. 5D depicts a cross-sectional front view of another embodiment of the cap **110'**. It is understood that FIGS. 5A and 5B may depict either the embodiment in 5C or the embodiment in 5D.

(58) In the arrangement illustrated in FIG. 5A, the cap **110** has an oval shaped perimeter. In other implementations, the perimeter of the cap **110** may take different forms (e.g., substantially polygonal, or substantially circular). The perimeter of the cap **110** has width C and a depth G. In one implementation, the width C is between 1.5 and 2.5 inches, such as 2 inches and the depth G is between 1 and 1.5 inches, such as 1.25 inches. The top of the cap **110** has a substantially flat surface with an oval shaped perimeter. The perimeter has a width J and a depth K. In one implementation, the width J is between 1.2 and 1.6 inches, such as 1.4 inches and the depth K is between 0.5 and 1 inches, such as 0.7 inches.

(59) FIG. 5B depicts the bottom view of the arrangement illustrated in FIG. 5A. The cap **110** has an outer wall **114**, an inner wall **116** extending within the outer wall **114**, and a seal **120** recessed in the inner wall **116**. The outer wall **114** defining a cavity **112** and the inner wall **116** defined by a post **118**.

(60) FIG. 5C depicts a cross-sectional front view of one embodiment of the cap **110**. In this embodiment, the post **118** extends into the cavity **112** and the seal **120** sits within the post **118**. In one example, a distal portion **122** of the post **118** extends past the seal **120**.

(61) FIG. 5D depicts a cross-sectional front view of a second embodiment of the cap **110'**. In this embodiment, the post **118'** extends into the cavity **112'** and the seal **120'** sits within the post **118'**. The seal **120'** extends to the extent of the post **118'** (e.g., the seal **120'** is at the distal end **122'** of the post **118'**) and the post **118'** extends to the extent of the cavity **112'**.

(62) In some embodiments, the outer wall **114**, the inner wall **116**, and the seal **120**, or any combination thereof, may be formed as a single piece. The single piece can be an integrated assembly of several parts or can be a unitary structure formed from a single piece of material. In some embodiments, the seal **120**, the outer wall **114**, and the inner wall **116** may be made from metal, plastic, rubber, or any suitable material. Any of these portions may be made from the same material or from different materials.

(63) As shown in FIG. 6A, the assembly **100** can be inverted and disposed over an eye **300** so that

the eye cup **140** is disposed over (e.g., in contact with) an eye socket **310** of the eye **300** (e.g., to that the eye cup **140** provides a supporting structure that supports the dispenser bottle **150** in an inverted position on the eye socket **310** of the user's eye **300**). The user can then squeeze the dispenser bottle **150** to dispense one or more drops into the eye **300**. This arrangement advantageously facilitates self-administration of a solution to a user's eye **300** and is user friendly as it does not require the user to assemble the eye cup onto the bottle and can apply it with one hand. Rather, the user simply removes the cap **110** from the dispenser bottle **150**, inverts the dispenser bottle **150**, places the eye cup **140** against the eye socket **310** of the eye **300**, and squeezes the dispenser bottle **150** to dispense the solution. Further, the nipple **130** is advantageously centered relative to the eye cup **140** (e.g., aligned with a central axis of the eye cup **140**) so that when the eye cup **140** is supported on the eye socket **310**, the nipple **130** is generally aligned (e.g., centered) relative to the eye **300**, further ensuring the dispensed solution will be delivered onto or over the eye **300**. Advantageously, the eye cup **140** is sized to dispose the distal edge **144** over (e.g., in contact with) the periorbital region **310** and align the nipple **130** with the center of the eye **300**. The distal edge **144** of the eye cup **140** is contoured to fit around the periorbital region **310**, advantageously allowing the eye cup **140** to comfortably sit or rest over the eye socket **310**. This positioning facilitates alignment of the nipple **130** with the eye **300** allowing for dispensing the eye drop(s) directly into the eye **300**. This positioning limits (e.g., prevents) waste from having the eye drops fall on the user's face, cheek, eyebrow, ear, etc. Though FIG. 6A depicts the use of the assembly **100**, the assembly **100'** can be used in the same manner and the advantages listed above also apply to the assembly **100'**. This is best illustrated in FIG. 6B.

(64) The contour of the distal edge **144** is defined by two axes. The first axis is defined by the nipple **130** such that the distal edge **144** is contoured circumferentially around the nipple **130**. The second axis is defined by a point in space approximating the centroid of the external curvature of a user's periorbital region **310**. The distal edge **144** is contoured relative to this axis such that the eye cup **140** fits against the user's face. The ability of the eye cup **140** to sit or rest over the periorbital region **310** advantageously facilitates use of the eye drop dispenser assembly **100** by a user with hand tremors, nervous disorders or children to dispense eye drops into the eye. Additionally, supporting (e.g., sitting or resting) the eye cup **140** over the eye socket **310** can inhibit (e.g., prevent) injury to the eye during delivery of eye drops and inhibit the nipple **130** from touching the user's eye **300**, thereby inhibiting (e.g., preventing) contamination of the eye **300**.

(65) FIGS. 7A and 8A show views of a dispenser assembly **100''** (hereafter "the assembly **100''**"), which is a version of the dispenser assembly **100'** designed for use with single-use vial cartridges. Some of the features of the assembly **100''** are similar to features of the assembly **100'** in FIGS. 1B, 3A-3C, 4B, 5D, and 8B. Thus, reference numerals used to designate the various components of the assembly **100''** are identical to those used for identifying the corresponding components of the assembly **100'** in FIGS. 1B, 3A-3C, 4B, 5D, and 8B, except that a "" has been added to the numerical identifier. Therefore, the structure and description for the various features of the assembly **100'** and how it is operated and controlled in FIGS. 1B, 3A-3C, 4B, 5D, and 8B are understood to also apply to the corresponding features of the assembly **100''**, except as described below. The dispenser assembly **100''** can house and deliver a solution (e.g., eye drops, eye wash, eye lid cleaner and/or wash) onto or over the eye or a portion thereof (e.g., eye lid).

(66) FIG. 7A depicts a cross-sectional front view of a third embodiment of the eye drop dispenser assembly **100''**. This is the single-use vial cartridge eye drop dispenser assembly **100''**. The dispenser assembly **100''** includes a dispenser bottle **150''** (e.g., dispenser body) with a base **152''**. Additionally, the cap **110'** described above can attach to the dispenser bottle **150''** in substantially the same way the cap **110'** attaches to the dispenser bottle **150'**. The cap **110'** can be removably couplable to an end of the dispenser bottle **150''** opposite to (e.g., distal from) the base **152''**. In some embodiments, the base **152''** is flat. In other embodiments, a portion, but less than all, of the base **152''** is flat. The assembly **100''** has a height A'' measured from the base **152''** to a top of the

cap **110'**. In one implementation, the height A' is between about 1.5 and 2.5 inches, such as 2 inches. In one example, the eye cup **140''** can be translucent. In one example, the eye cup **140''** can be transparent.

(67) The section is taken at a midpoint of the depth G'' of the assembly **100''** (see FIG. 8A). The dispenser bottle **150''** has a chamber **180''** that can hold a single-use vial cartridge **170''**. FIGS. 7A, 8A shows a single-use vial cartridge **170''** inserted into the chamber **180''**. The single-use vial cartridge **170''** is (removably) inserted into the chamber **180''** from the bottom of the dispenser bottle **150''** (e.g., the end of the dispenser bottle **150''** opposite the eye cup **140''**) and it is held in the chamber **180''** (e.g., by friction, such as via a press-fit connection). In some embodiments, the single-use vial cartridge **170''** can be a single-use vial. In some embodiments the single-use vial cartridge **170''** can be a single-use eye drop vial. The chamber **180''** has an opening **182''** positioned within the surface **164''**. The opening **182''** can include a retention lip **184''** which can hold the single-use vial cartridge **170''** in place. The dispenser bottle **150''** can be squeezed to dispense fluid from the single-use vial cartridge **170''**. For example, flexing the dispenser bottle **150''** can cause the chamber **180''** flex and press on the single-use vial cartridge **170''**, thereby releasing fluid from the single-use vial cartridge **170''**. The retention lip **184''** can be a snap-fit which allows the single-use vial cartridge **170''** to be inserted or removed but retains the single-use vial cartridge **170''** in its position while in use. In another implementation, the retention lip **184''** can be resilient (e.g., act as a spring to allow insertion of the single-use vial cartridge **170''** into the chamber **180''** and removal of the single-use vial cartridge **170''** from the chamber **180''**). The nipple **130''** is integral to the single-use vial cartridge **170''**. The nipple **130''** can extend out of the surface **164''** (e.g., out of the opening **182''**). The surface **164''** is between the second surface **160''** and the base **152''**. The positioning of the surface **164''** creates a recess cavity **156''** around the nipple **130''**. The recess cavity **156''** extends into the dispenser bottle **150''** and away from the distal edge **144''** of the eye cup **140''**. The positioning of the recess cavity **156''** allows the nipple **130''** to be recessed proximal (e.g., the nipple **130''** is nested in the recess cavity **156''** of the dispenser bottle **150''**) to the dispenser bottle **150''** while projecting towards the distal edge **144''** of the eye cup **140''**, allowing the dispenser assembly **100''** to be more compact than the dispenser assembly **100**. The eye cup **140''** fits within a cavity **112'** of the cap **110'** and the circumferential wall **142''** defines an oval shape surrounding the nipple **130''**. The nipple **130''** is disposed between distal edge **144''** of the circumferential wall **142''** and the surface **164''** of the dispenser bottle **150''**. In one example, the circumferential wall **142'** is partially disposed within the recess cavity **156''**.

(68) Similar to the description above with regard to FIG. 3C, the outer wall **114'** can also be disposed around the shoulder **154''** of the bottle **150''**. The outer wall **114'** can be tangential to the outer surface of the dispenser bottle **150''**. The seal **120'** can sit adjacent the top of the nipple **130''** and presses against the nipple **130''** to inhibit leakage of a solution or liquid from the nipple **130''**. In one example, the seal **120'** can have a shape similar to (e.g., a mirror image of) a shape of the nipple **130''**). The post **118'** is disposed within the eye cup **140''** when the cap **110'** is connected to the dispenser bottle **150''**.

(69) In the illustrated arrangement, the bottle **150''** has a height B'' measured from the base **152''** to the second surface **160''**. In one implementation, the height B'' is between 1 and 1.75 inches, such as 1.25 inches or 1.5 inches. The assembly **100''** has a height F'' (for the eye cup **140''**) measured from the surface **164''** to the distal edge **144''**. In one example, the eye cup **140''** extends from the second surface **160''** (e.g., the eye cup **140''** is outside the recess cavity **156''**). In another example, the eye cup **140''** extends from the surface **164''** so that at least a portion of the circumferential wall **142''** of the eye cup **140''** is in the recess cavity **156''** of the bottle **150''**. In one implementation, the height F'' is between 0.25 and 0.75 inches, such as 0.4 inches or 0.6 inches. The assembly **100''** also has a height D''. Height D'' is the height of the cap **110'**. The height D'' is greater than height F''. In one implementation, the height D'' is between 0.5 and 1 inches, such as 0.75 inches or 0.9 inches. The dispenser bottle **150''** has a height E'' measured from the second surface **160''** to the surface

164" (e.g., of the recess cavity **156"**). In one implementation, the height **E"** is between 0.1 and 0.6 inches, such as 0.25 inches or 0.4 inches. In some arrangements, the second surface **160"** may be coplanar with the surface **164"** and height **E"** is zero inches. In some examples, the second surface **160"** and the surface **164"** can be portions of the same surface. The single-use dispenser bottle **150"** has a height **I"** measured from the nipple **130"** to the second surface **160"**. In one implementation the height **I"** is between 0" and 0.1 inches, such as 0.05 inches.

(70) FIG. 7B illustrates a front section view of the eye drop dispenser assembly **100'**. As illustrated, the dispenser bottle **150'** has a height **I'** measured from the nipple **130'** to the second surface **160'**. In one implementation the height **I'** is between 0 and 0.1 inches, such as 0.05 inches. As discussed above regarding FIG. 3C, the bottle **150'** has a height **B'** measured from the base **152'** to the second surface **160'**. In one implementation, the height **B'** is between 1 and 1.75 inches, such as 1.25 inches or 1.5 inches. The assembly **100'** has a height **F'** (for the eye cup **140'**) measured from the surface **164'** to the distal edge **144'**. In one implementation, the height **F'** is between 0.25 and 0.75 inches, such as 0.4 inches or 0.5 inches. The assembly **100'** also has a height **D'**. Height **D'** is the height of the cap **110'**. In one implementation, the height **D'** is between 0.5 and 1 inches, such as 0.75 inches or 0.9 inches. The dispenser bottle **150'** has a height **E'** measured from the second surface **160'** to the surface **164'** (e.g., the recessed cavity of the dispenser bottle). In one implementation, the height **E'** is between 0.1 and 0.6 inches, such as 0.25 inches or 0.4 inches. The overall height **A'** from the bottom of the dispenser bottle **150'** to the top of the cap **110'** is the sum of heights **B'** and **D'** or between 1.5 inches and 2.75 inches.

(71) FIG. 8A illustrates a top view of the eye drop dispenser assembly **100"** of FIG. 7A. The dispenser bottle **150"** is shown with the distal edge **144"** of the eye cup **140"** having an oval shape and disposed around the nipple **130"**. In this embodiment, the dispenser bottle **150"**, has an oval shaped perimeter. The nipple **130"** is centered relative to the eye cup **140"** and the dispenser bottle **150"**. In other implementations, the perimeter of the dispenser bottle **150"** or eye cup **140"** may take different forms (e.g. substantially polygonal, or substantially circular). In this embodiment, the single-use vial cartridge **170"** is held in place by a retention lip **184"**.

(72) As illustrated, the dispenser bottle **150"** has a depth **G"** and the eye cup **140"** has a depth **L"**. In one implementation, the depth **L"** is less than the depth **G"**. In one implementation, the depth **G"** is between 1.25 inches and 1.75 inches, such as 1.4 inches. In one implementation, the depth **L"** is between 1.0 inches and 1.5 inches such as 1.25 inches. As illustrated, the dispenser bottle **150"** has a width **C"** and the eye cup **140"** has a width **N"**. In one implementation, the width **C"** is greater than the width **N"**. In one implementation, the width **C"** is between 1.5 and 2.5 inches, such as 1.9 inches. In one implementation, the width **N"** is between 1.4 inches and 1.8 inches, such as 1.6 inches.

(73) FIG. 8B illustrates a top view of the eye drop dispenser assembly **100'** of FIG. 1B. As illustrated, the dispenser bottle **150'** has a depth **G'** and the eye cup **140'** has a depth **L'**. In one implementation, the depth **L'** is less than the depth **G'**. In one implementation, the depth **G'** is between 1 and 1.5 inches, such as 1.2 inches or 1.4 inches. In one implementation, the depth **L'** is between 1.0 inches and 1.5 inches such as 1.25 inches. As illustrated the dispenser bottle **150'** has a width **C'** and the eye cup **140'** has a width **N'**. In one implementation, the width **N'** is less than the width **C'**. In one implementation, the width **C'** between is 1.5 and 2.5 inches, such as 1.9 inches. In one implementation, the width **N'** is between 1.4 inches and 1.8 inches, such as 1.6 inches.

(74) FIG. 8C illustrates a bottom view of the dispenser assembly **100"**. As illustrated, the base **152"** of the dispenser bottle **150"** has an opening **186"** which opens into the chamber **180"**. The single-use vial cartridge **170"** can be inserted through the opening **186"** and into the chamber **180"** of the dispenser bottle **150"**.

(75) The assembly **100'** differs from the assembly **100** in that the dispenser assembly **100'** has a smaller external envelope. This is accomplished by recessing the nipple **130'** as depicted in FIG. 3C. This allows height **D'** to be less than height **D** and height **A'** to be less than height **A**. Bottle **150**

can hold a larger volume of fluid than bottle **150'**.

(76) The assembly **100''** differs from the assembly **100** in that the dispenser assembly **100''** has a smaller external envelope. This is accomplished by recessing the nipple **130''** as depicted in FIG. 7A. This allows height D'' to be less than height D and height A'' to be less than height A.

(77) The assembly **100, 100'** is advantageously easy to use (e.g., it is easy to remove the cap **110, 110'**, position the eye cup **140, 140'** over the eye **300**, and dispense drops in the optimal location without contaminating the nipple **130, 130'**). For example, because the eye cup **140, 140'** is integrated (e.g., fixed) with the dispenser bottle **150, 150'**, the user can simply remove the cap **110, 110'**, invert the dispenser bottle **150, 150'** and position it over (e.g., in contact with) their eye socket **310** and squeeze the dispenser bottle **150, 150'** to dispense the solution from the dispenser bottle **150, 150'**. Additionally, the fixed eye cup **140, 140'** advantageously reduces the likelihood that the eye cup **140, 140'** will be lost, used improperly, or that the nipple **130, 130'** will be or get contaminated.

(78) The assembly **100''** is advantageously easy to use (e.g., it is easy to remove the cap **110'**, insert a single-use vial cartridge **170''**, remove the top of the single-use vial cartridge **170''**, position the eye cup **140''** over the eye **300**, and dispense drops in the optimal location without contaminating the nipple **130''**) or touching the nipple **130''** to the eye **300**. This facilitates safe, user-friendly, self-administration of eye drops to a user's eye **300**. For example, because the eye cup **140''** is integrated (e.g., fixed) with the dispenser bottle **150''**, the user can simply remove the cap **110, 110'**, invert the dispenser bottle **150''** and position it over (e.g., in contact with) their eye socket **310** and squeeze the dispenser bottle **150''** to dispense the solution from the dispenser bottle **150''**, thereby facilitating self-administration of the eye solution by a user without needing assistance from someone else.

(79) FIGS. 9A-15B illustrate a fourth embodiment of an eye drop dispenser assembly **200** (the "assembly **200**"). The assembly **200** includes a detachable and invertible eye cup **230**, a dispenser bottle **210**, and a cap **250**. FIG. 9A shows an assembled front view, of the assembly **200** and FIG. 9B shows an exploded front view of the assembly **200**. FIG. 9C shows a front cross section view of the assembly **200** assembled with the eye cup **230** in the stowed position so that the assembly **200** is generally shaped like a water droplet or tear drop. The cap **250** can be a single piece (e.g., monolithic, seamless) with a sidewall **253**. The dispenser bottle **210**, the eye cup **230** and/or the cap **250** can optionally be made in part or whole of plastic, silicone or rubber. However, the dispenser bottle **210**, the eye cup **230** and/or the cap **250** can be made of other suitable materials (e.g., flexible or resilient materials). The cap **250** is removably couplable to the dispenser bottle **210** and can optionally have a conical shape. In one example, the eye cup **230** can be translucent. In one example, the eye cup **230** can be transparent.

(80) As shown in FIG. 10, the dispenser bottle **210** has an internal chamber **219** that can removably receive and hold a single-use vial cartridge **270** (e.g., eye drop cartridge), so the assembly **200** is a single-use vial cartridge eye drop dispenser assembly. FIGS. 9C-13 shows a single-use vial cartridge **270** inserted into the chamber **219**, with FIGS. 10 and 12 showing a shorter single-use vial cartridge **270** that entirely resides in the chamber **219** and FIGS. 9C and 11 showing a longer single-use vial cartridge **270** that extends out of the bottom of the dispenser bottle **210**. In one example, the eye drops dispensed by the eye drop dispenser assembly **200** can be eye lubricant drops. The liquid can be a formulation for treating a condition of the eyes (e.g., dry eye, bacterial infections, cataracts, glaucoma, or for treating eye injuries). The dispenser bottle **210** can optionally be made of plastic, silicone, rubber, or other suitable resilient or flexible material. The dispenser bottle **210** is a single piece (e.g., a monolithic, seamless single piece). The dispenser bottle **210** can optionally include a lower portion **212** with a sidewall **213** and an upper portion **214** with a sidewall **215**, the dispenser bottle **210** having a shoulder **216** (e.g., lower shoulder) between the upper portion **214** and the lower portion **212**. In some implementations, the lower portion **212** can be cylindrical. As best shown in FIGS. 9C-10, the shoulder **216** can have a curved or contoured

edge **216A** on the front, side and rear of the dispenser bottle **210**. In one implementation, the sidewall **213** is curved or contoured. In one implementation, the sidewall **215** is generally linear and tapers between the shoulder **216** and a second shoulder **217** (e.g., upper shoulder **217**), as shown in FIG. **10**. That is, the lower portion **212** has a larger width than the upper portion **214** (e.g., to hold the eye cup).

(81) The dispenser bottle **210** includes a locking mechanism **220** disposed between (e.g., radially between) the second shoulder **217** and the nipple **272**. The cap **250** and/or the eye cup **230** can each attach to the dispenser bottle **210** with a locking mechanism **220**. In one example, the locking mechanism **220** is a snap-fit. The locking mechanism **220** can additionally or alternatively include one or more threads for threadably coupling with the cap **250** and/or the eye cup **230**. Other suitable locking mechanisms can be used. For example, the locking mechanism can be a clip-on mechanism (e.g., that allows the eye cup **230** to clip onto the dispenser bottle **210**, such as via multiple clips that clip into depressions in the bottle dispenser), or a press-fit mechanism, where the eye cup **230** is press-fit (e.g., friction fit) onto the dispenser bottle **210**. In another example, the locking mechanism **220** can be a threaded coupling, where the eye cup **230** is threadably coupled to threads on the dispenser bottle **210**. In another example, the locking mechanism **220** includes one or more recesses (e.g., multiple recesses, two recesses, four recesses) and one or more hooks (e.g., multiple hooks, two hooks, four hooks).

(82) The eye cup **230**, seen in FIG. **10**, can have a first edge **231** (e.g., bottom edge), a second edge **232** (e.g., top edge) opposite the first edge **231**, and a sidewall **233** that extends between and interconnects the first edge **231** and the second edge **232**. The eye cup **230** can optionally be transparent. The eye cup **230** can optionally be made of plastic, silicone, rubber or other suitable resilient or flexible material. The eye cup **230** can be a single piece (e.g., a monolithic, seamless single piece). In other embodiments the eye cup **230** can be multiple pieces that can be assembled together. The eye cup **230** can be hollow with the first edge **231** defining a first opening **O1** and the second edge **232** defining a second opening **O2** opposite the first opening **O1**. The eye cup **230** is removably couplable to the dispenser bottle **210**. The eye cup **230** can have one or more protrusions or tabs (e.g., multiple, two, four spaced apart protrusions) that extend radially inward from the second edge **232** into the second opening **O2**. The first opening **O1** and first edge **231** can have a larger perimeter than the second opening **O2** and second edge **232**, the sidewall **233** tapering between the first edge **231** and the second edge **232** (e.g., tapering in a curved manner). As best shown in FIG. **13**, the first edge **231** (and first opening **O1**) can have an oval shape and the second edge **232** (and second opening **O2**) can have a circular shape. With continued reference to FIG. **10**, the first edge **231** can be curved. In one implementation, the first edge **231** is curved or contoured and generally corresponds with a curvature of the contoured edge **216A** of the shoulder **216**. The second edge **232** can be linear.

(83) With continued reference to FIG. **9A**, **9C**, when the eye cup **230** is in the stowed position on the dispenser bottle **210**, the eye cup **230** extends around the upper portion **214** (see FIG. **10**) of the dispenser bottle **210** (and completely below the nipple **272**). In the stowed position (shown e.g. in FIG. **9A**, **9C**), the sidewall **233** of the eye cup **230** aligns (e.g., defines a generally continuous edge) with the sidewall **213** of the lower portion **212** of the dispenser bottle **210**, and the sidewall **233** also generally aligns with the sidewall **253** of the cap **250**, advantageously providing a compact assembly when the eye cup **230** is in the stowed position over the dispenser bottle **210**.

(84) With continued reference to FIG. **10**, when the eye cup **230** is in the deployed position on the dispenser bottle **210**, the eye cup **230** extends around the nipple **272** (and completely above the upper portion **214** of the dispenser bottle **210**). In the deployed position, the eye cup **230** is oriented so that the second edge **232** is adjacent the shoulder **217**, and coupled to the locking mechanism **220** (as described above), and the first edge **231** and first opening **O1** are disposed above the nipple **272** (e.g., so that the first edge **231** defines a topmost end of the assembly **200**). In the deployed position, the eye cup **230** can be placed adjacent the eye socket **310** to align the nipple **272** with the

eye **300**, allowing the user to deliver (e.g., self-administer) solution to the eye **300**.

(85) In some embodiments, the single-use vial cartridge **270** can be a single-use vial cartridge. In some embodiments the single-use vial cartridge **270** can be a single-use eye drop vial cartridge. The chamber **219** can include a retention lip **284** which can hold the single-use vial cartridge **270** in place. The dispenser bottle **210** can be squeezed to dispense fluid from the single-use vial cartridge **270**. For example, flexing the dispenser bottle **210** can cause the chamber **219** flex and press on the single-use vial cartridge **270**, thereby releasing fluid from the single-use vial cartridge **270**. The retention lip **284** can be a snap-fit which allows the single-use vial cartridge **270** to be inserted or removed but retains the single-use vial cartridge **270** in its position while in use. In another implementation, the retention lip **284** can be resilient (e.g., act as a spring to allow insertion of the single-use vial cartridge **270** into the chamber **219** and removal of the single-use vial cartridge **270** from the chamber **219**). As illustrated in FIG. **14**, the bottom surface of the dispenser bottle **210** has an opening **218**. The single-use vial cartridge **270** can be inserted into the opening **218**. The nipple **272** is integral to the single-use vial cartridge **270**. The nipple **272** can extend out past the retention lip **284**.

(86) FIGS. **11-14** show schematic views of the eye drop dispenser assembly **200**. FIG. **11** shows a front view of the eye drop dispenser assembly **200** with the eye cup **230** in the stowed position. FIG. **12** shows a front view of the eye drop dispenser assembly **200** with the eye cup **230** in the deployed position. FIG. **13** shows a top view of the eye drop dispenser assembly **200** with the eye cup **230** in the deployed position and with the cap **250** removed. FIG. **14** shows a bottom view of the eye drop dispenser assembly **200**.

(87) As illustrated in FIG. **11**, the assembly **200** has a height **H1** between a base of the dispenser bottle **210** and a top of the cap **250**, and a height **H2** between the base of the dispenser bottle **210** and the end of the nipple **272**. The assembly **200** has a height **H6** between a top edge of the dispenser bottle **210** and the base of the dispenser bottle **210**. In one implementation the height **H2** and the height **H6** are the same height. In another implementation the height **H2** is taller than the height **H6**. In one implementation, the height **H1** is between about 2.5 and 3.5 inches, such as 3 inches, and the height **H2** is between about 2.3 and about 3.4 inches, such as 2.9 inches. In one implementation, the height **H6** is between about 2.3 and about 3.4 inches, such as 2.9 inches. The eye cup **230** has a height **H3** of between about 0.75 inches and about 1.3 inches, such as 1 inch. As illustrated in FIG. **13** the dispenser bottle **210** has a width **W1** of between about 1.3 inches and 2.3 inches, such as 1.8 inches, and a depth **D1** of between about 1 inch and about 1.5 inches, such as 1.25 inches. The cap **250** can have a width **W2** of between about 0.75 inches and about 1.25 inches, such as about 1 inch, and a height **H5** of between about 0.75 inches and about 1.25 inches, such as about 1 inch. A gap **H4** between the second edge **232** of the eye cup **230** and the bottom of the cap **250** can be between about 0.005 inches and about 0.07 inches, such as about 0.05 inches. When in the deployed or inverted position, the eye cup **230** can extend a height **H5** past the nipple **272**. In one example, illustrated in FIG. **12**, the height **H5** is between about 0.1 inches and about 0.6 inches, such as about 0.25 inch and facilitates a soft landing of the eye drop onto the eye. The height **H5** can also help ensure the eye drop is delivered into the eye and does not run off the eye (e.g., onto the user's face, such as cheek).

(88) FIG. **15A** illustrates the insertion of a single-use vial cartridge **270** into the assembly **200**. FIG. **15B** illustrates the eye drop dispenser **200** in an inverted position over an eye **300** to deliver an eye drop thereto. As shown in FIG. **15B**, the assembly **200** can be inverted and disposed over an eye **300** so that the eye cup **230** is disposed over (e.g., in contact with) an eye socket **310** of the eye **300** (e.g., to that the eye cup **230** provides a supporting structure that supports the dispenser bottle **210** in an inverted position on the eye socket **310** of the user's eye **300**). The user can then squeeze the dispenser bottle **210** to dispense one or more drops into the eye **300**. This arrangement advantageously facilitates self-administration of a solution to a user's eye **300**. The user simply removes the cap **250** from the dispenser bottle **210**, attaches the eye cup **230** to the dispenser bottle

210, removes the top or tab of the single-use vial cartridge **270**, inserts the single-use vial cartridge **270** into the dispenser bottle **210** (e.g., as shown FIG. **15A**), inverts the dispenser bottle **210**, places the eye cup **230** against the eye socket **310** of the eye **300**, and squeezes the dispenser bottle **210** to dispense the solution.

(89) The assembly **200** is advantageously compact because of the removable eye cup **230**.

Additionally, the assembly **200** is advantageously easy to use (e.g., it is easy to remove the cap **250**, remove the eye cup **230** from the stored position and reattach the cap **250** in the deployed (inverted) position, remove the top or tab of a single-use vial cartridge **270**, insert the single-use vial cartridge **270**, position the eye cup **230** over the eye **300**, and dispense the eye solution (e.g., eye drops, eye wash, eye lid wash) in the optimal location without contaminating the nipple **272** or touching the nipple **272** to the eye **300**. This facilitates safe, user-friendly, self-administration of eye solution to a user's eye **300**. The assembly **200** advantageously facilitates self-administration of the eye solution by a user without needing assistance from someone else.

(90) FIG. **16A** illustrates the insertion of a single-use vial cartridge **170''** into the assembly **100''**.

FIG. **16B** illustrates the dispenser assembly **100''** in an inverted position over an eye **300** to deliver an eye drop thereto. FIG. **16C** shows a cross-sectional view of the assembly **100''** in FIG. **16A** with the single-use vial cartridge **170''** (removably) inserted therein. As shown in FIG. **16B**, the assembly **100''** can be inverted and disposed over an eye **300** so that the eye cup **140''** is disposed over (e.g., in contact with) an eye socket **310** of the eye **300** (e.g., so that the eye cup **140''** provides a supporting structure that supports the dispenser bottle **150''** in an inverted position on the eye socket **310** of the user's eye **300**). The user can then squeeze the dispenser bottle **150''** to dispense the eye solution (e.g., one or more drops) into or over the eye **300**. This arrangement advantageously facilitates self-administration of a solution to a user's eye **300** and is user friendly as it does not require the user to assemble the eye cup onto the bottle and can apply it with one hand. Rather, the user simply removes the cap **110''** from the dispenser bottle **150''**, removes the top or tab of the single-use vial cartridge **170''**, inserts the single-use vial cartridge **170''** into the dispenser bottle **150''** (e.g., as shown FIG. **16A**), inverts the dispenser bottle **150''**, places the eye cup **140''** against the eye socket **310** of the eye **300**, and squeezes the dispenser bottle **150''** to dispense the solution.

(91) FIG. **17** illustrates a single-use vial cartridge **470**. Each feature of the single-use vial cartridge **470** applies to the single-use vial cartridges **170''** and **270** described herein. The single-use vial cartridge **470** includes a nipple **472** and a cover tab **474**. The nipple **472** is fluidically attached to a fluid reservoir **476**. Below the fluid reservoir **476** is an insertion tab **478**. The insertion tab **478** can be used to position the single-use vial cartridge **470** within a dispenser bottle. The insertion tab **478** can also be used to remove the single-use vial cartridge **470** from the dispenser bottle. The cover tab **474** can be removed to dispense fluid (e.g., eye drops, eye wash, eye lid cleaner and/or wash) from the fluid reservoir **476** through the nipple **472**. The single-use vial cartridge **470** can optionally be made in part or whole of plastic (e.g., polypropylene).

(92) FIG. **18A** illustrates a schematic front cross-sectional view of a dispenser assembly **100''**. FIG. **18B** illustrates a schematic side cross-sectional view of a dispenser assembly **100''** (e.g., a view offset 90 degrees from the view in FIG. **18A**). Some of the features of the assembly **100''** are similar to features described with reference to FIGS. **7A** and **8A**. Therefore, the structure and description for the various features of the assembly **100''** and how it is operated and controlled in FIGS. **7A** and **8A** are understood to also apply to the corresponding features of the assembly **100''** illustrated in FIGS. **18A** and **18B**, and vice versa, except as described below. The dispenser assembly **100''** can removably or releasably house a cartridge vial (not shown) to deliver a solution (e.g., eye drops, eye wash, eye lid cleaner and/or wash) onto or over the eye or a portion thereof (e.g., eye lid).

(93) The dispenser assembly **100''** includes a dispenser bottle **150''** (e.g., dispenser body) with a base **152''**. Additionally, the cap **110''** described above can attach to the dispenser bottle **150''** in

substantially the same way the cap **110'** attaches to the dispenser bottle **150'**. The cap **110"** can be removably couplable to an end of the dispenser bottle **150"** opposite to (e.g., distal from) the base **152"**. In some embodiments, the base **152"** is flat. In other embodiments, a portion, but less than all, of the base **152"** is flat.

(94) The cross-section illustrated in FIG. **18A** is taken at a midpoint of the depth **G"** of the assembly **100"** (see FIG. **8A**). The dispenser bottle **150"** has a chamber **180"** (cartridge chamber) that can releasably and removably hold a single-use vial cartridge **170"** (not pictured). The single-use vial cartridge **170"** is (removably) inserted into the chamber **180"** from the bottom of the dispenser bottle **150"** (e.g., the end of the dispenser bottle **150"** opposite the eye cup **140"**) via an opening **185"** and it is held in the chamber **180"** by a retention element **189"**. The retention element **189"** is a circumferential ridge which extends into the chamber **180"** to create a friction fit between the retention element **189"** and the vial cartridge **170"**. In some embodiments, the retention element **189"** can be an annular snap-fit. The chamber **180"** has an opening **182"** defined within the surface **164"**. The opening **182"** can include a retention lip **184"** which can aid in holding the single-use vial cartridge **170"** in place, providing an axial limit for the body of the vial cartridge **170"**. The retention lip **184"** can be a snap-fit which allows the single-use vial cartridge **170"** to be inserted or removed but retains the single-use vial cartridge **170"** in its position while in use. In another implementation, the retention lip **184"** can be resilient (e.g., act as a spring to allow insertion of the single-use vial cartridge **170"** into the chamber **180"** and removal of the single-use vial cartridge **170"** from the chamber **180"**), such as a resilient tapered wall. The retention lip **184"** can in one example extend circumferentially about a central axis of the dispenser bottle **150"**. In another example, the retention lip **184"** can be two angled or tapered walls on opposite sides of the central axis of the dispenser bottle **150"**. The dispenser bottle **150"** can be squeezed to dispense fluid from the single-use vial cartridge **170"**. As illustrated, the chamber **180"** is defined by a frame that extends from the opening **185"** to the opening **182"** and includes side wall portions **183"** positioned along two sides of the chamber **180"**. A pair of arms **188"** can extend outward from the chamber **180"** towards the walls of the dispenser bottle **150"** so that the arms **188"** are integral to the chamber **180"** and are positioned between the chamber **180"** and the dispenser bottle **150"**. The arms **188"** are positioned along two sides of the chamber **180"** between the side wall portions **183"** so that the chamber **180"** has a rectangular cross section. The arms **188"** can be attached to living hinges **181"** (e.g., resilient walls) which connect to the frame of the chamber **180"** proximal to the opening **185"**. As illustrated in FIG. **18B**, a first arm **188"** can extend towards the front wall of the dispenser bottle **150"** and a second arm **188"** can extend towards the rear wall of the dispenser bottle **150"**. In some embodiments, the arms **188"** can extend toward the side walls of the dispenser bottle **150"**. The arms **188"** can pivot on the living hinges **181"** into the chamber **180"** and towards the central axis of the dispenser bottle **150"** (e.g., the arms **188"** can move relative to the frame of the chamber **180"**). The arms **188"** can transfer a force or pressure applied to the dispenser bottle **150"** by a user when squeezing the dispenser bottle **150"** to the single-use vial cartridge **170"**. For example, flexing the dispenser bottle **150"** can push on the arms **188"**, causing the chamber **180"** to press on the single-use vial cartridge **170"** and compress the single-use vial cartridge **170"**, thereby releasing fluid from the single-use vial cartridge **170"** via the nipple of the single-use vial cartridge **170"**.

(95) In some embodiments, the chamber **180"**, arms **188"**, and eye cup **140"** are integral and together form a single body (e.g., the chamber **180"**, arms **188"**, and eye cup **140"** form a single seamless body or are molded simultaneously). In some embodiments, the chamber **180"** and arms **188"** are integral and together form a single body. In some embodiments, the eye cup **140"** can attach to the dispenser bottle **150"** via a snap-fit connection or a threaded attachment. In some embodiments, the chamber **180"** can attach to the dispenser bottle **150"** via a snap-fit connection or a threaded attachment. In some embodiments, the eye cup **140"** can attach to the chamber **180"**. The eye cup **140"** can attach to the chamber **180"** via a snap-fit connection or a threaded

attachment.

(96) FIG. 19A illustrates a schematic front cross-sectional view of a dispenser assembly 200. FIG. 19B illustrates a schematic side cross-sectional view of a dispenser assembly 200 (e.g., a view offset 90 degrees from the view in FIG. 18A). Some of the features of the assembly 200 are similar to features described with reference to FIGS. 9A-12. Therefore, the structure and description for the various features of the assembly 200 and how it is operated and controlled in FIGS. 9A-12 are understood to also apply to the corresponding features of the assembly 200 illustrated in FIGS. 19A and 19B, and vice versa, except as described below. The dispenser assembly 200 can removably or releasably house a cartridge vial (not shown) to deliver a solution (e.g., eye drops, eye wash, eye lid cleaner and/or wash) onto or over the eye or a portion thereof (e.g., eye lid).

(97) The dispenser bottle 210 has a chamber 219 that can hold a single-use vial cartridge 270 (not pictured). The single-use vial cartridge 270 can be (removably) inserted into the chamber 219 from an opening 218 in the bottom of the dispenser bottle 210 (e.g., the end of the dispenser bottle 210 opposite the eye cup 230) and it is held in the chamber 219 by a retention element 262. The retention element 262 is a circumferential ridge which extends into the chamber 219 to create a friction fit between the retention element 262 and the vial cartridge 270. In some embodiments, the retention element 262 can be an annular snap-fit. The chamber 219 has an opening 282 at the top of the dispenser bottle 210 (e.g., the end of the dispenser bottle 210 where the eye cup 230 attaches to the dispenser bottle 210). The opening 282 can include a retention lip 284 which can hold the single-use vial cartridge 270 in place. The retention lip 284 can be a snap-fit which allows the single-use vial cartridge 270 to be inserted or removed but retains the single-use vial cartridge 270 in its position while in use. In another implementation, the retention lip 284 can be resilient (e.g., act as a spring to allow insertion of the single-use vial cartridge 270 into the chamber 219 and removal of the single-use vial cartridge 270 from the chamber 219). The dispenser bottle 210 can be squeezed to dispense fluid from the single-use vial cartridge 270. As illustrated, the chamber 219 is defined by a frame that extends from the opening 218 to the opening 282 and includes side wall portions 266 positioned at along two sides of the chamber 219. A pair of arms 260 can extend outward from the chamber 219 towards the walls of the dispenser bottle 210 so that the arms 260 are positioned between the chamber 219 and the dispenser bottle 210. The arms 260 are positioned along two sides of the chamber 219 between the side wall portions 266. The arms 260 can be attached to living hinges 264 (e.g., resilient walls) which connect to the frame of the chamber 219 proximal to the opening 218. As illustrated, a first arm 260 can extend towards the front wall of the dispenser bottle 210 and a second arm 260 can extend towards the rear wall of the dispenser bottle 210. In some embodiments, the arms 260 can extend towards the side walls of the dispenser bottle 210. The arms 260 can pivot on the living hinges 264 into the chamber 219 and towards the central axis of the dispenser bottle 210 (e.g., the arms 260 can move relative to the frame of the chamber 219). The arms 260 can transfer a force or pressure applied to the dispenser bottle 210 to the single-use vial cartridge 270. For example, flexing the dispenser bottle 210 can push on the arms 260 causing the chamber 219 to compress and press on the single-use vial cartridge 270, thereby releasing fluid from via the nipple of the single-use vial cartridge 270.

(98) In some embodiments, the chamber 219 and arms 260 are integral and together form a single body seamless body. In some embodiments, the chamber 219 can attach to the dispenser bottle 210 via a snap-fit connection or a threaded attachment. In some embodiments, the eye cup 230 can attach to the chamber 219. The eye cup 230 can attach to the chamber 219 via a snap-fit connection or a threaded attachment. In some embodiments, the cap 250 can attach to the chamber 219 near the retention lip 284 via a snap-fit connection or a threaded attachment.

(99) While certain embodiments of the inventions have been described, these embodiments have been presented by way of example only and are not intended to limit the scope of the disclosure. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the systems and methods

described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure. Accordingly, the scope of the present inventions is defined only by reference to the appended claims.

(100) Features, materials, characteristics, or groups described in conjunction with a particular aspect, embodiment, or example are to be understood to be applicable to any other aspect, embodiment or example described in this section or elsewhere in this specification unless incompatible therewith. All of the features disclosed in this specification (including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. The protection is not restricted to the details of any foregoing embodiments. The protection extends to any novel one, or any novel combination, of the features disclosed in this specification (including any accompanying claims, abstract and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed.

(101) Furthermore, certain features that are described in this disclosure in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations, one or more features from a claimed combination can, in some cases, be excised from the combination, and the combination may be claimed as a subcombination or variation of a subcombination.

(102) Moreover, while operations may be depicted in the drawings or described in the specification in a particular order, such operations need not be performed in the particular order shown or in sequential order, or that all operations be performed, to achieve desirable results. Other operations that are not depicted or described can be incorporated in the example methods and processes. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the described operations. Further, the operations may be rearranged or reordered in other implementations. Those skilled in the art will appreciate that in some embodiments, the actual steps taken in the processes illustrated and/or disclosed may differ from those shown in the figures. Depending on the embodiment, certain of the steps described above may be removed, others may be added. Furthermore, the features and attributes of the specific embodiments disclosed above may be combined in different ways to form additional embodiments, all of which fall within the scope of the present disclosure. Also, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described components and systems can generally be integrated together in a single product or packaged into multiple products.

(103) For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. Not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those skilled in the art will recognize that the disclosure may be embodied or carried out in a manner that achieves one advantage or a group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

(104) Conditional language, such as “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements, and/or steps are included or are to be performed in any particular embodiment.

(105) Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require the presence of at least one of X, at least one of Y, and at least one of Z.

(106) Language of degree used herein, such as the terms “approximately,” “about,” “generally,” and “substantially” as used herein represent a value, amount, or characteristic close to the stated value, amount, or characteristic that still performs a desired function or achieves a desired result. For example, the terms “approximately,” “about,” “generally,” and “substantially” may refer to an amount that is within less than 10% of, within less than 5% of, within less than 1% of, within less than 0.1% of, and within less than 0.01% of the stated amount. As another example, in certain embodiments, the terms “generally parallel” and “substantially parallel” refer to a value, amount, or characteristic that departs from exactly parallel by less than or equal to 15 degrees, 10 degrees, 5 degrees, 3 degrees, 1 degree, or 0.1 degree.

(107) The scope of the present disclosure is not intended to be limited by the specific disclosures of preferred embodiments in this section or elsewhere in this specification and may be defined by claims as presented in this section or elsewhere in this specification or as presented in the future. The language of the claims is to be interpreted broadly based on the language employed in the claims and not limited to the examples described in the present specification or during the prosecution of the application, which examples are to be construed as non-exclusive.

(108) Of course, the foregoing description is that of certain features, aspects and advantages of the present invention, to which various changes and modifications can be made without departing from the spirit and scope of the present invention. Moreover, the devices described herein need not feature all of the objects, advantages, features and aspects discussed above. Thus, for example, those of skill in the art will recognize that the invention can be embodied or carried out in a manner that achieves or optimizes one advantage or a group of advantages as taught herein without necessarily achieving other objects or advantages as may be taught or suggested herein. In addition, while a number of variations of the invention have been shown and described in detail, other modifications and methods of use, which are within the scope of this invention, will be readily apparent to those of skill in the art based upon this disclosure. It is contemplated that various combinations or subcombinations of these specific features and aspects of embodiments may be made and still fall within the scope of the invention. Accordingly, it should be understood that various features and aspects of the disclosed embodiments can be combined with or substituted for one another in order to form varying modes of the discussed devices.

Claims

1. A dispenser assembly, comprising: a dispenser bottle comprising: a circumferential sidewall; a first surface; and a chamber extending from an opening in the first surface to an opening at a bottom of the dispenser bottle, the chamber having a chamber wall spaced inward from the circumferential sidewall to define a gap between the chamber wall and the circumferential sidewall, the chamber configured to removably receive and hold a vial therein; one or more arms positioned between the chamber and an outer surface of the dispenser bottle, the one or more arms movable towards a central axis of the chamber when the outer surface of the dispenser bottle is pressed; and an eye cup fixedly coupled to a second surface of the dispenser bottle, the eye cup comprising: a circumferential sidewall disposed around the chamber; and a distal edge of the circumferential sidewall defining an oval shape, the distal edge configured to be disposed over an eye socket to facilitate delivery of a solution from the vial into a user's eye.
2. The dispenser assembly of claim 1, further comprising a cap removably couplable to the dispenser bottle over the chamber.

3. The dispenser assembly of claim 2, wherein the dispenser bottle further comprises a shoulder and the cap is removably couplable to the shoulder.
 4. The dispenser assembly of claim 2, wherein an outer surface of the cap is tangential to the dispenser bottle.
 5. The dispenser assembly of claim 1, wherein the dispenser bottle has a recess cavity extending away from the distal edge of the eye cup.
 6. The dispenser assembly of claim 5, wherein the circumferential sidewall of the eye cup is partially disposed within the recess cavity.
 7. The dispenser assembly of claim 1, wherein the distal edge of the eye cup is contoured.
 8. The dispenser assembly of claim 1, further comprising a retention lip extending from the chamber, the retention lip configured to retain a vial in the chamber.
 9. The dispenser assembly of claim 8, wherein the retention lip extends from the first surface.
 10. The dispenser assembly of claim 9, wherein the retention lip is a snap-fit.
 11. The dispenser assembly of claim 9, wherein the retention lip is configured to act as a spring to allow insertion of a single-use vial cartridge into the chamber and removal of the single-use vial cartridge from the chamber.
 12. The dispenser assembly of claim 1, wherein a maximum width and depth of the eye cup is smaller than a maximum width and depth of the dispenser bottle.
 13. The dispenser assembly of claim 1, wherein chamber has a rectangular cross-section.
 14. The dispenser assembly of claim 1, further comprising a cap removably couplable to the dispenser bottle over a nipple, the cap comprising: an outer wall; an inner wall extending within the outer wall; and a seal at or proximal an end of the inner wall.
 15. The dispenser assembly of claim 1, wherein the circumferential sidewall of the dispenser bottle is configured to flex and press on a single-use vial disposed in the chamber.
 16. A dispenser assembly, comprising: a dispenser bottle having: a chamber extending from an opening in a first surface to an opening at a bottom of the dispenser bottle, the chamber having a chamber wall spaced inward from a circumferential sidewall of the dispenser bottle to define a gap between the chamber wall and the circumferential sidewall; a coupling mechanism proximate the chamber; and one or more arms positioned between the chamber and an outer surface of the dispenser bottle, the one or more arms movable toward a central axis of the chamber when the outer surface of the dispenser bottle is pressed; and an eye cup having a first open end and a second open end opposite the first open end and configured to removably fit over at least a portion of the dispenser bottle, the first open end having an oval shape and the second open end having a circular shape, the eye cup being couplable to the coupling mechanism of the dispenser bottle in a stowed position where the eye cup is disposed about an upper portion of the dispenser bottle, the eye cup further being couplable to the coupling mechanism of the dispenser bottle in a deployed position where the eye cup is disposed so that it extends around and distally from the coupling mechanism.
 17. The dispenser assembly of claim 16, wherein the dispenser bottle further comprises an upper portion recessed relative to a lower portion, and wherein an outer surface of the eye cup circumferentially aligns with an outer surface of the lower portion of the dispenser bottle when in the stowed position.
 18. The dispenser assembly of claim 16, wherein the dispenser bottle has an upper portion between a lower portion and the coupling mechanism, the lower portion having a greater width than the upper portion, the upper portion tapering from the lower portion.
 19. The dispenser assembly of claim 16, further comprising a cap configured to be couplable to the coupling mechanism.
 20. The dispenser assembly of claim 19, wherein the cap is configured to be couplable to the coupling mechanism while the eye cup is in the stowed position.
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